TYPES OF VARIABLES

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Variables

- In statistical research, a variable is defined as an attribute of an object of study
- Choosing which variables to measure is central to good experimental design.

Example

If you want to test whether some plant species are more salt-tolerant than others, some key variables you might measure include the **amount of salt** you add to the water, the **species of plants** being studied, and variables related to plant health like **growth** and **wilting**.

You need to know which types of variables you are working with in order to choose appropriate statistical tests and interpret the results of your study.

Source: https://www.scribbr.com/methodology/types-of-variables/

Type of Variable ?

1. What type of data does the variable contain?

2. What part of the experiment does the variable represent?

Types of data:

- Data is a specific measurement of a variable – it is the value you record in your datasheet.
- Data is generally divided into two categories:

Quantitative data represents amounts

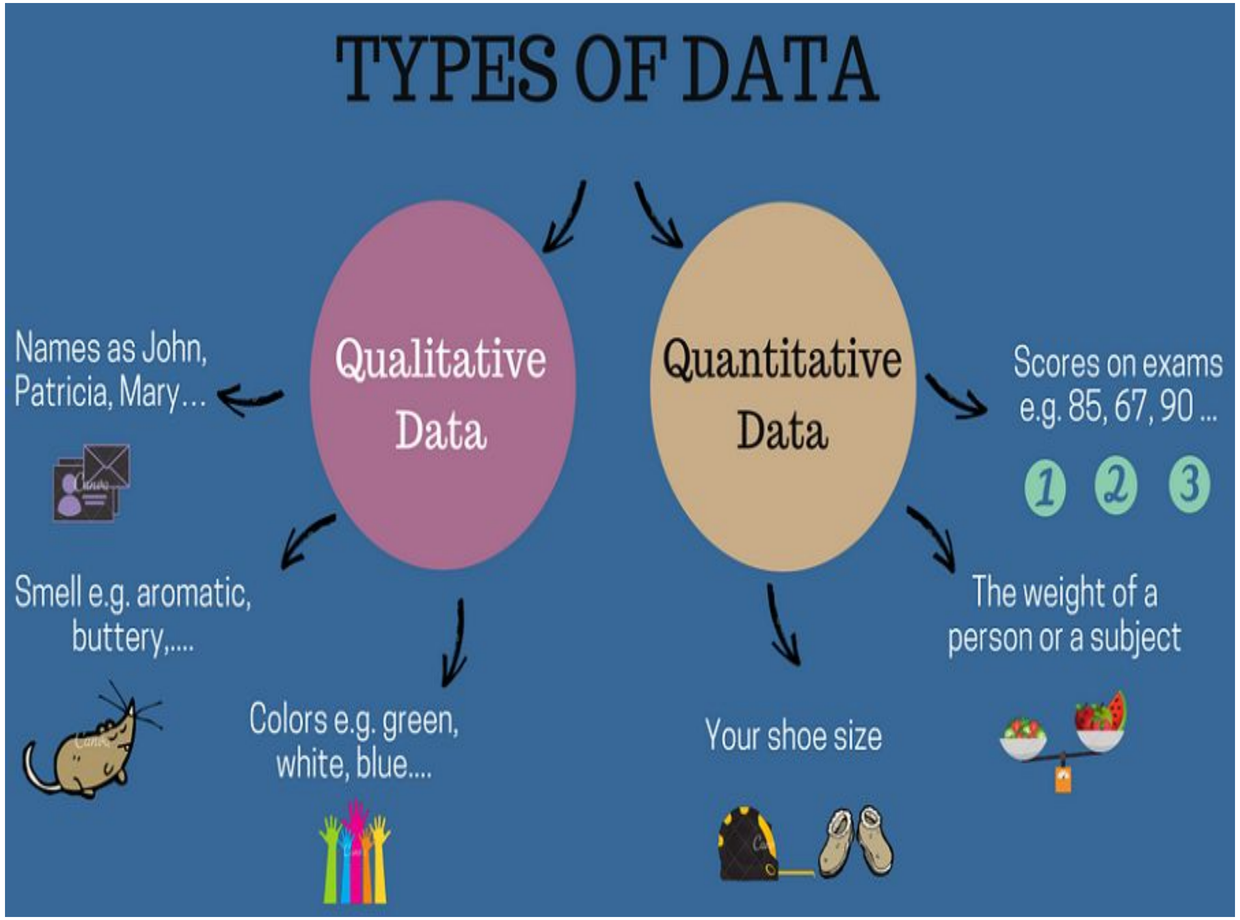
Categorical data represents groupings

Types of Data

A variable that contains quantitative data is a **quantitative variable**;

A variable that contains categorical data is a **categorical variable**.

Each of these types of variables can be broken down into further types.



Quantitative Data/ Numerical Data

- Quantitative data is data that can be expressed as a number or can be quantified. In other words, quantitative data can be measured by numerical variables.

- Quantitative data are easily amenable to statistical manipulation and can be represented with a wide variety of statistical types of graphs and charts such as line, graph, bar graph, scatter plot, box and whisker plot, and etc.

Examples of quantitative data:

- Scores on tests and exams e.g. 85, 67, 90 and etc.
- The weight of a person or a subject.
- The number of hours of study.
- Your shoe size.
- The square feet of an apartment.
- The temperature in a room.
- The volume of a gas and etc.

Types of quantitative data:

There are two types of quantitative variables:

1. **discrete**
2. **continuous**.

Discrete vs continuous variables

- **Discrete data** – a count that involves integers. Only a limited number of values is possible. The discrete values cannot be subdivided into parts. For example, the number of children in a school is discrete data. You can count whole individuals. You can't count 1.5 kids.
- **Continuous data** – information that could be meaningfully divided into finer levels. It can be measured on a scale or continuum and can have almost any numeric value. For example, you can measure your height at very precise scales — meters, centimeters, millimeters and etc.

Type of variable	What does the data represent?	Examples
Discrete variables (aka integer variables)	Counts of individual items or values.	•Number of students in a class •Number of different tree species in a forest
Continuous variables (aka ratio variables)	Measurements of continuous or non-finite values.	•Distance •Volume •Age

Categorical Data/ Qualitative Data

- Qualitative data, also known as the categorical data, describes the data that fits into the categories. Qualitative data are not numerical. The categorical information involves categorical variables that describe the features such as a person's gender, home town etc. Categorical measures are defined in terms of natural language specifications, but not in terms of numbers.
- Sometimes categorical data can hold numerical values (quantitative value), but those values do not have a mathematical sense. Examples of the categorical data are birthdate, favourite sport, school postcode. Here, the birthdate and school postcode hold the quantitative value, but it does not give numerical meaning.

Categorical Variables-Types

- There are three types of categorical variables: **binary**, **nominal**, and **ordinal variables**.

Type of variable	What does the data represent?	Examples
Binary variables (aka dichotomous variables)	Yes or no outcomes.	•Heads/tails in a coin flip •Win/lose in a football game
Nominal variables	Groups with no rank or order between them.	•Species names •Colors •Brands
Ordinal variables	Groups that are ranked in a specific order.	•Finishing place in a race •Rating scale responses in a survey

Example

- To keep track of your salt-tolerance experiment, you make a data sheet where you record information about the variables in the experiment, like salt addition and plant health.
- To gather information about plant responses over time, you can fill out the same data sheet every few days until the end of the experiment.

Data Sheet// Salt Tolerance// Date: _______________

Sample #	Plant Species	Salt Added (mg/L water)	Starting Height (cm)	Growth (cm) (current height – starting height)	Wilting (rank 0-10)	Survival (1=survived, 0=died)
1	A	0	12			
2	A	100	13			
3	A	250	11			
4	B	0	25			
5	B	100	26			
6	B	250	25			

Nominal continuous ordinal binary.

Parts of the experiment: Independent vs dependent variables

- Experiments are usually designed to find out what effect one variable has on another
- You manipulate the **independent variable** (the one you think might be the **cause**) and then measure the **dependent variable** (the one you think might be the **effect**) to find out what this effect might be.
- You will probably also have variables that you hold constant (**control variables**) in order to focus on your experimental treatment.

Independent vs dependent vs control variables

Type of variable	Definition	Example (salt tolerance experiment)
Independent variables (aka treatment variables)	Variables you manipulate in order to affect the outcome of an experiment.	The amount of salt added to each plant's water.
Dependent variables (aka response variables)	Variables that represent the outcome of the experiment.	Any measurement of plant health and growth: in this case, plant height and wilting.
Control variables	Variables that are held constant throughout the experiment.	The temperature and light in the room the plants are kept in, and the volume of water given to each plant.

Example data sheet

Data Sheet// Salt Tolerance// Date: ________________

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Blue colour: Independent Variables
Violet colour: Dependent Variables

Data Sets

Total Examples: 14, Positive Examples: 9, Negative Examples: 5

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	yes
D6	Rain	cool	Normal	Strong	Yes
D7	Overcast	Cool	High	Weak	No
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Other common types of variables

Type of variable	Definition	Example (salt tolerance experiment)
Confounding variables	A variable that hides the true effect of another variable in your experiment. This can happen when another variable is closely related to a variable you are interested in, but you haven't controlled it in your experiment.	Pot size and soil type might affect plant survival as much or more than salt additions. In an experiment you would control these potential confounders by holding them constant.
Latent variables	A variable that can't be directly measured, but that you represent via a proxy.	Salt tolerance in plants cannot be measured directly, but can be inferred from measurements of plant health in our salt-addition experiment.
Composite variables	A variable that is made by combining multiple variables in an experiment. These variables are created when you analyze data, not when you measure it.	The three plant health variables could be combined into a single plant-health score to make it easier to present your findings.

Levels of Measurement

- **Level of measurement refers to how precisely a variable has been measured**.
- In Statistics, the variables or numbers are defined and categorized using different **scales of measurements**
  - Scale: A scale is a device or an object used to measure or quantify any event or another object.
- Each **level of measurement** scale has specific properties that determine the various uses of statistical analysis

Levels of Measurements

There are four different scales of measurement.

- Nominal Scale
- Ordinal Scale
- Interval Scale
- Ratio Scale

Nominal Scale

Nominal level	Examples of nominal scales
You can categorize your data by labelling them in mutually exclusive groups, but there is no order between the categories.	•City of birth •Gender •Ethnicity •Car brands •Marital status

Ordinal Level

Ordinal level	Examples of ordinal scales
You can categorize and rank your data in an order, but you cannot say anything about the intervals between the rankings. Although you can rank the top 5 Olympic medallists, this scale does not tell you how close or far apart they are in number of wins.	•Top 5 Olympic medallists •Language ability (e.g., beginner, intermediate, fluent) •Likert-type questions (e.g., very dissatisfied to very satisfied)

Interval Data

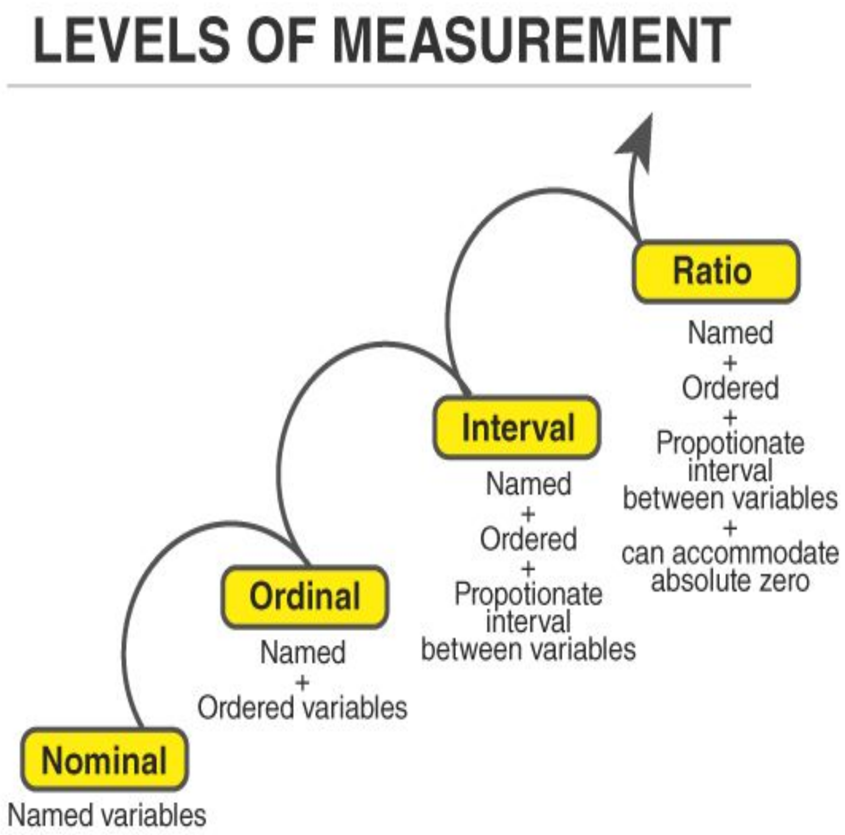
Interval level	Examples of interval scales
You can categorize, rank, and infer equal intervals between neighboring data points, but there is no true zero point. The difference between any two adjacent temperatures is the same: one degree. But zero degrees is defined differently depending on the scale – it doesn't mean an absolute absence of temperature. The same is true for test scores and personality inventories. A zero on a test is arbitrary; it does not mean that the test-taker has an absolute lack of the trait being measured.	•Test scores (e.g., IQ or exams) •Personality inventories •Temperature in Fahrenheit or Celsius

Stanford–Binet Fifth Edition (SB5) classification[42][48]

IQ Range ("deviation IQ")	IQ Classification
140+	Very gifted or highly advanced
130–140	Gifted or very advanced
120–129	Superior
110–119	High average
90–109	Average
80–89	Low average
70–79	Borderline impaired or delayed
55–69	Mildly impaired or delayed

Ratio level	Examples of ratio scales
You can categorize, rank, and infer equal intervals between neighboring data points, and there is a true zero point. A true zero means there is an absence of the variable of interest. In ratio scales, zero does mean an absolute lack of the variable.	•Height •Age •Weight •Temperature in Kelvin

Central Tendency



Geometric Mean, Hormonic Mean

Arithmetic Mean

Median

Mode

- **Correlation** means there is a statistical association between variables.
- **Causation** means that a change in one variable causes a change in another variable